

LESSON 106 (0.5 GROUND and 1.5 AATD)

Lesson Objective

During this lesson, the student will learn how to taxi and practice straight and level flight techniques in the training device. The student will also review the effects of controls. The student will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the landing gear, hydraulic, fuel, and oil systems.
 - Quiz the student on the readings as needed.
 - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic flight maneuvers.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
 - Section 07 - Description and operation of the airplane and its systems.

Lesson Content (Ground)

Taxiing Procedures

- Introduce the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Straight and Level Flight Techniques

- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.

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- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain heading.
- Utilize the Altimeter and VSI to maintain altitude.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic when weather permits.
- FAR regulation 91.111 - operating near other aircraft.
 - Cannot operate near another aircraft to create a collision hazard.
 - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
 - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
 - During taxi.
 - Before climbs.
 - During descents.
 - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Lesson Content (AATD)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all of the checklists.
- Assist if needed.
- Have the student go through the "passenger briefing" and the "engine fire during start brief".

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- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Review Effects of Controls as Appropriate

- If needed, refer to the lesson plan for 104.

Straight and Level Flight at Cruise Speed

- Demonstrate entering and maintaining straight and level flight.
- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain pitch.
- Utilize the Altimeter and VSI to maintain altitude.

Straight and Level Flight at Selected Airspeeds

- For changes greater than 20 knots:
 - Full power or idle power.
 - Anticipate by 5 knots.
- For changes less than 20 knots:
 - 100 RPM change = 5 knots change.

Straight and Level Flight with Flaps

Flap Settings / Effects	Lift	Nose	Stabilator(initially)	Drag	Airspeed	Stabilator (due to drag)
0 - 10 Degrees	↑↑↑	↑↑	Forward	↑	↓	Backward
10 - 25 Degrees	↑↑	↑↑	Forward	↑↑	↓↓	Backward
25 - 40 Degrees	↑	↑	Forward	↑↑↑	↓↓↓	Backward
40 - 25 Degrees	↓	↓	Backward	↓↓↓	↑↑↑	Forward
25 - 10 Degrees	↓↓	↓↓	Backward	↓↓	↑↑	Forward
10 - 0 Degrees	↓↓↓	↓↓	Backward	↓	↑	Forward

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 107.
- Assign homework:

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- Read chapter 7 of the Airplane Flying Handbook - Ground Reference Maneuvers:
 - Drift and Ground Track Control.

Completion Standards

This lesson is complete when the student can:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Have a weather briefing and a completed TOLD card prepared.
3. Demonstrate proper sequential checklist procedures for before and after start, before and after taxi, run-up and takeoff.
4. Demonstrate an understanding of straight and level flight techniques in the training device.
5. Explain the relationship between power, airspeed, and angle of attack at various speeds.